SERIES AND PARALLEL CIRCUITS

1. In a series circuit the supply voltage equals the _____of the voltage drops around the circuit. (A)
2. In a series circuit the current has only one _____ to follow. (A)
3. In a series circuit the current is _____ in every part of the circuit. (A)
4. In a parallel circuit the current from the power supply is equal to the _____ of the currents in the branches. (A)
5. In a parallel circuit the voltage drops across parallel components is _____(A)
6. Car headlights are wired in parallel. One advantage of this is (M)_____

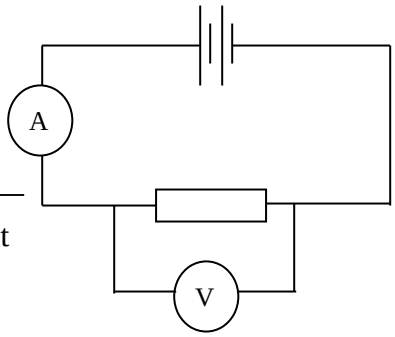
   _____
7. A light bulb is connected across a battery. A second light bulb is then connected in series with the first. How does this affect the brightness of the first bulb? (M)_____

   _____

   Now the two light bulbs are rearranged so they are in parallel. How does this affect the brightness of the first bulb?(M)_____

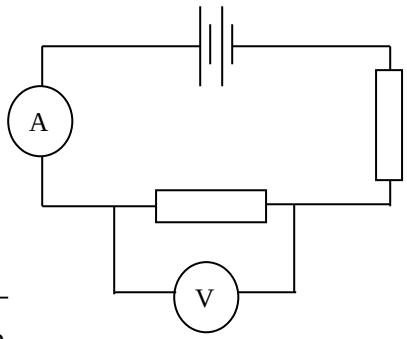
   _____

8. In the circuit diagram shown;



   (a) Is this circuit series or parallel? (A)_____
   (b) How is the Voltmeter connected? (A)_____
   (c) Draw an arrow on the wire showing which way the current is flowing. (A)
   (d) If the battery is 3V what is the voltage reading? (A) _____
   (e) If the resistor is 6 Ohms what is the current reading? (M) _____

9. A second 6 Ohm resistor is added to the circuit as shown.

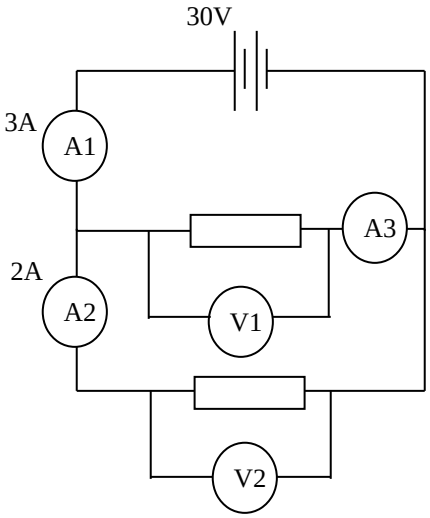


   (a) What is the total resistance of the circuit? (A) _____
   (b) What is the new voltmeter reading?(M)_____
   (c) What is the new ammeter reading? (M)_____
   (d) Is this a series or parallel circuit? (A) _____
   (e) If one of the resistors burned out, what would happen to the current from the battery? (A)_____

(f) If the ammeter was shifted to between the two resistors (both going)what current value would it read? (A)_____

(g) If the voltmeter was shifted to be across the other resistor what voltage value would it read? (A) _____

10. In the circuit shown:



(a) Is this a series or parallel circuit?(A) _____

(b) What is the reading on (A) A3 _____

V1 _____

V2 _____

(c) Are the two resistors the same value?(A) _____

Explain your answer? (M)_____

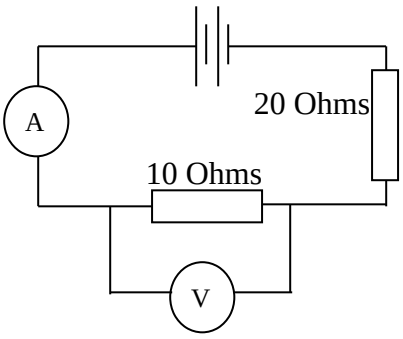
_____

_____

(d) If the resistors were in fact light bulbs and one of the light bulbs blew out would the other light bulb go out?(A) _____

Would the reading on A1 go up, or down, or stay the same?(M)_____

11. The circuit shown has two resistors. One is 10 Ohms and the other is 20 Ohms. The voltmeter reads 4 Volts.



(a) Calculate the current in the 10 Ohm resistor. (M)

(b) Calculate the voltage across the 20 Ohm resistor. (M)

(c) Calculate the voltage of the battery. (A)

12. Draw a circuit diagram which has a battery, a switch, a rheostat and a light bulb in series. Add an ammeter to measure the current and a voltmeter to measure the voltage across the light bulb. (M)